ASSIGNMENT 3

Q9 - JSON Audit

JavaScript solution

```
// q9.js
"use strict";

const rawData = [
  '{"user":"Alex","age":25}',
  '{"id":2}',
  '{invalid}',
  '{"user":"Mina","age":"22"}'
];

const clean = [];
const errors = [];
const under18 = [];

for (let index = 0; index < rawData.length; index++) {
  const lineNumber = index + 1;
  try {
    const record = JSON.parse(rawData[index]);
    if (record.user === undefined || record.age === undefined) {
      throw new Error("Missing required key: user or age");
    }

    const age = Number(record.age);
    if (!Number.isFinite(age)) {
      throw new Error("Age is not a valid number");
    }

    const normalized = { user: String(record.user), age };
    clean.push(normalized);
    if (age < 18) under18.push(normalized);
  } catch (error) {
    errors.push({ lineNumber, message: error.message });
    console.log(`Line ${lineNumber}: ${error.message}`);
  }
}

console.log("Clean records:", clean);
console.log("Errors:", errors);
console.log("Under-18 users:", under18);
```

Explanation / Debug notes

JSON.parse throws for malformed JSON and control jumps to catch. Valid JSON is then checked for required keys. Age is normalized to a number. The provided data contains Alex and Mina as clean records; the record with only id is missing required keys, and {invalid} is invalid JSON. The under-18 list is empty for this sample.

Run the code in the browser console or Node.js. Save each solution in its own .js file as required.